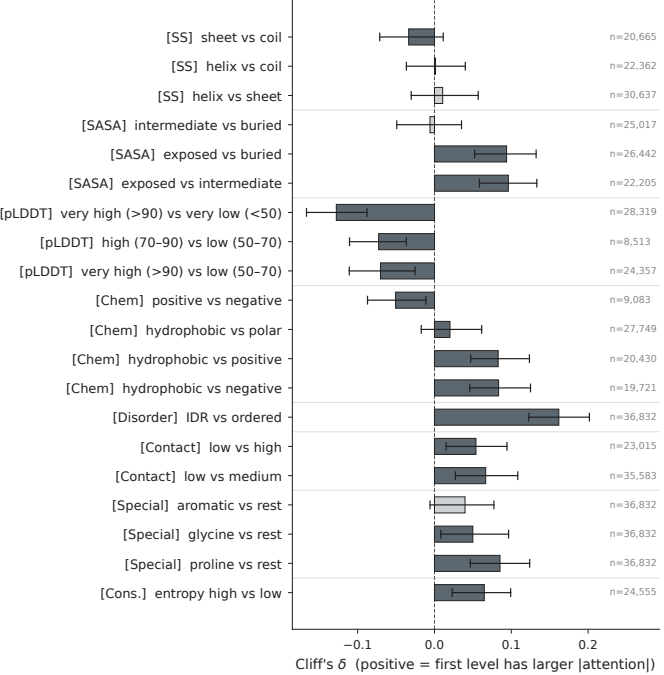
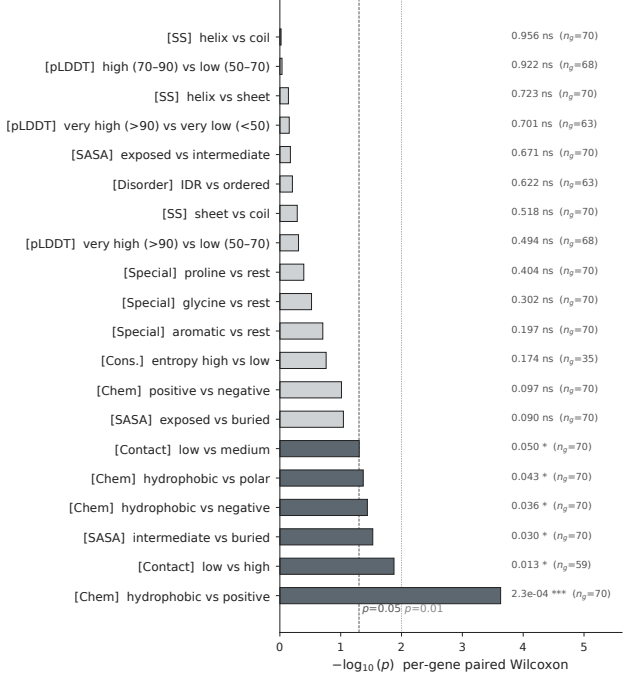
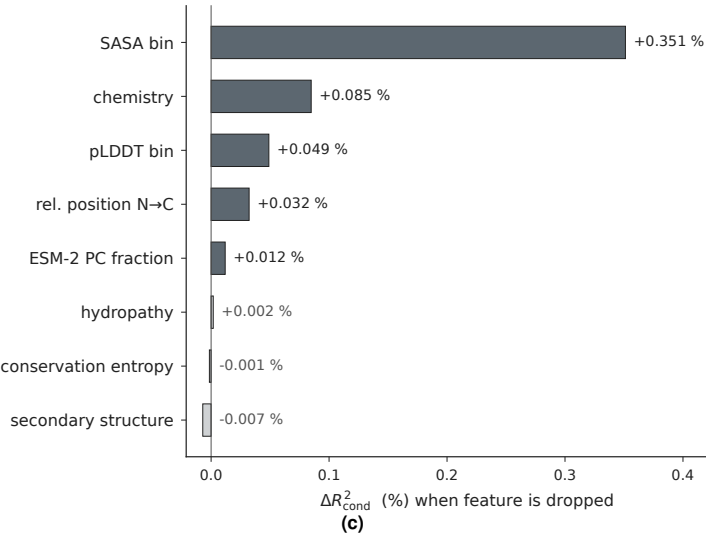

[SS] sheet vs coil
[SS] helix vs coil
[SS] helix vs sheet
[SASA] intermediate vs buried
[SASA] exposed vs buried
[SASA] exposed vs intermediate
[pLDDT] very high (>90) vs very low (<50)
[pLDDT] high (70–90) vs low (50–70)
[pLDDT] very high (>90) vs low (50–70)
[Chem] positive vs negative
[Chem] hydrophobic vs polar
[Chem] hydrophobic vs positive
[Chem] hydrophobic vs negative
[Disorder] IDR vs ordered
[Contact] low vs high
[Contact] low vs medium
[Special] aromatic vs rest
[Special] glycine vs rest
[Special] proline vs rest
[Cons.] entropy high vs low
n=20,665
n=22,362
n=30,637
n=25,017
n=26,442
n=22,205
n=28,319
n=8,513
n=24,357
n=9,083
n=27,749
n=20,430
n=19,721
n=36,832
n=23,015
n=35,583
n=36,832
n=36,832
n=36,832
n=24,555
−0.1
0.0
0.1
0.2
Cliff's δ (positive = first level has larger |attention|)

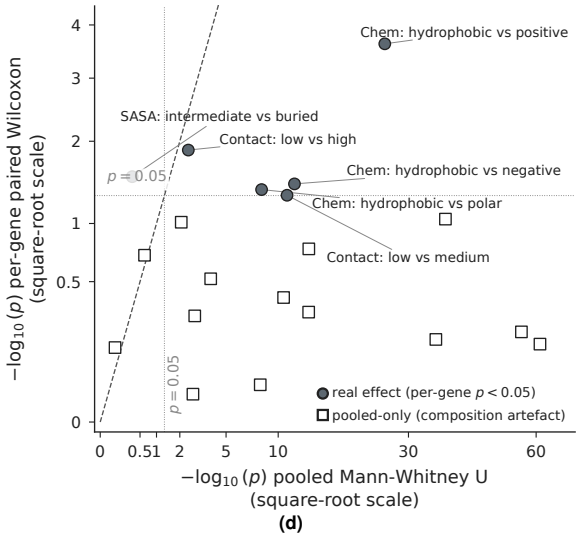
(a)


[SS] helix vs coil
[pLDDT] high (70–90) vs low (50–70)
[SS] helix vs sheet
[pLDDT] very high (>90) vs very low (<50)
[SASA] exposed vs intermediate
[Disorder] IDR vs ordered
[SS] sheet vs coil
[pLDDT] very high (>90) vs low (50–70)
[Special] proline vs rest
[Special] glycine vs rest
[Special] aromatic vs rest
[Cons.] entropy high vs low
[Chem] positive vs negative
[SASA] exposed vs buried
[Contact] low vs medium
[Chem] hydrophobic vs polar
[Chem] hydrophobic vs negative
[SASA] intermediate vs buried
[Contact] low vs high
[Chem] hydrophobic vs positive
0.956 ns (n_g=70)
0.922 ns (n_g=68)
0.723 ns (n_g=70)
0.701 ns (n_g=63)
0.671 ns (n_g=70)
0.622 ns (n_g=63)
0.518 ns (n_g=70)
0.494 ns (n_g=68)
0.404 ns (n_g=70)
0.302 ns (n_g=70)
0.197 ns (n_g=70)
0.174 ns (n_g=35)
0.097 ns (n_g=70)
0.090 ns (n_g=70)
0.050 * (n_g=70)
0.043 * (n_g=70)
0.036 * (n_g=70)
0.030 * (n_g=70)
0.013 * (n_g=59)
2.3e-04 *** (n_g=70)
p=0.05 p=0.01
0
1
2
3
4
5
$-\log_{10}(p)$ per-gene paired Wilcoxon

(b)


SASA bin
chemistry
pLDDT bin
rel. position N→C
ESM-2 PC fraction
hydropathy
conservation entropy
secondary structure
+0.351 %
+0.085 %
+0.049 %
+0.032 %
+0.012 %
+0.002 %
-0.001 %
-0.007 %
0.0
0.1
0.2
0.3
0.4
ΔR^2_{cond} (%) when feature is dropped

(c)


Chem: hydrophobic vs positive
SASA: intermediate vs buried
Contact: low vs high
p = 0.05
Chem: hydrophobic vs negative
Chem: hydrophobic vs polar
Contact: low vs medium
p = 0.05
real effect (per-gene $p < 0.05$)
pooled-only (composition artefact)
0
0.5
1
2
3
4
0
0.51
2
5
10
30
60
$-\log_{10}(p)$ per-gene paired Wilcoxon (square-root scale)
$-\log_{10}(p)$ pooled Mann-Whitney U (square-root scale)

(d)